R2 · RESEARCHER GUIDE

Propose a simulation

Say precisely what a simulation should let a student do, so it can be built.

For physics staff

This is for the person who knows the physics, not the person who writes the code. You do not need to be able to build a simulation to get one into AIPLA — you need to be able to say precisely what it should let a student *do*. Drafting it in an AI chat is encouraged, and there is a prompt below that makes the draft land in the right shape.

What a simulation is here

A simulation in AIPLA is a small interactive bench that opens beside the tutor in the student's workspace. The student changes something, watches what happens, and records readings. The tutor sees those readings and can ask about them.

Nine are live today:

Simulation	The student's job
Boldkast	Launch a projectile, find the angle for maximum range
LED Planck	Measure LED threshold voltages, fit Planck's constant
KineBot	Read motion graphs
Elkedel	Heat water, measure energy in and temperature out, find the efficiency

Simulation	The student's job
Faseovergange	Heat ice to steam, read the plateaus off the heating curve
Bølgefart	Set f and λ, work out the wave speed
Interferens	Add two waves, find constructive, destructive and beats
Sol, Jord og Måne	Change viewpoint, scale and time; solve five missions on day, seasons, moon phases, eclipses and scale
Hældningslab	Type in the functions, read points off the graphs, draw lines through two points and find the slope at a point (mathematics)

Note the shape they share. **The simulation never shows the answer.** The kettle reports energy and temperature and never the efficiency; the wave bench reports f, λ and T and never the speed. That withholding is the exercise, and it is the single most useful thing you can specify when you propose one.

The one question worth getting right

> **What does the student measure, and what do they work out from it?**

If you can answer that in one sentence, the rest follows. If the answer is "they watch it", it is a diagram rather than a simulation, and a picture in the activity will serve better.

Two things follow from your answer:

- **The readings the bench must show.** These are instrument faces — a thermometer, a stopwatch, a frequency readout. Showing them is fine.
- **The value that must stay hidden.** The thing being calculated. It never appears on screen, and it goes into a private note for the tutor instead so the tutor can mark

an answer without handing it over.

What belongs in the simulation, and what does not

The bench holds the live experiment: the thing being simulated, the controls that act on it, the instruments reading it, and a small table of the readings the student has captured.

Everything else is better placed elsewhere, and placing it elsewhere makes it *more* useful, not less:

If your draft has	Where it actually goes
A numbered task list, "your turn" steps	the tutor, which can adapt them to the student
Constants, a formula sheet	the tutor, on request
A long problem statement	the activity's goal, which the teacher writes
A quiz, a multiple-choice check	an activity element
Its own chat box or "ask AI" button	deleted — AIPLA supplies the tutor
Its own title bar and logo	deleted — the app draws that

The bench is about 700 px wide on a laptop and 390 px on a phone, so this is not tidiness. Anything that is not the experiment is crowding the experiment out.

How to draft one

Use an AI chat, and paste the authoring prompt. It is published at https://aipla.ku.dk/project/build-a-simulation (https://aipla.ku.dk/project/build-a-simulation) (copy button), and as a plain file your chat can fetch by URL: https://aipla.ku.dk/sim-authoring-prompt.txt (https://aipla.ku.dk/sim-authoring-prompt.t

xt). It tells the AI the size limit, the "no external resources" rule, the light-on-white visual standard and the event vocabulary, so what comes back can be dropped in with little rework rather than rebuilt.

> **Tip**
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> **Start from the prompt even if you have built simulations with an AI before.** Two things come with it that are easy to miss without it. The tutor is told when a student takes a measurement or finishes a mission, so it responds on its own instead of waiting for the teacher to prompt it. And the control panels fold away and start folded on small screens, so they do not cover the experiment. A Sun–Earth–Moon simulation built from a blank chat in September 2026 worked well in class and still needed both added afterwards.

Then describe your experiment. A good brief says:

1. **The physics.** The relationship the student should come away with.
2. **What they control**, with ranges and units.
3. **What the instruments read.**
4. **What must stay hidden**, explicitly.
5. **What "done" looks like** — the capture or commit action.
6. **What the tutor may do to it**, if anything — for example "jump to the next full moon" or "switch to the view from the Moon". See below.

Open the result in a browser and try it. You are the physics reviewer; nobody downstream can catch a wrong model, and the automated checks certainly cannot.

Letting the tutor act on the simulation

By default the tutor only *watches* the bench. It can also *act* on it — move the clock, switch the view — if the simulation declares those actions as **commands** in its catalogue entry. Each command has a name, the values it takes, the sentence the student sees ("Sprang til næste fuldmåne"), and a power level: **view** (changes what is shown; the student can change it straight back), **scaffold** (puts a task or message on the student's screen) or **restrict** (locks something). How far a tutor may go is set by its teaching approach; with nothing set it may use only **view** commands. Every change shows the student a card in the chat. *Sol, Jord og Måne* is the first bench that does this.

If you want this, list the actions in your brief, in plain words, and say which ones would take something away from the student. Leave it out and the bench is watched, not driven.

What happens to your draft

Hand over the HTML file. It then goes through a documented path (`mcp-app-artefact`) that does the following, none of which is your problem:

- splits anything that turned out to be two experiments into two benches;
- strips the task list into the tutor's private brief;
- rewrites the text so Danish and English come from one place;
- wires the capture button so the tutor actually receives the readings and the student sees a "shared with the AI" card;
- runs the security, size, layout and phone-width gates;
- drives it in a real browser and captures the events;
- writes the tutor's private note, including your reference values;
- declares any tutor commands you asked for in the catalogue entry, each with its power level.

Then it is in the library and any teacher can attach it to an activity.

Two things we will ask you

Sign off the physics. ADR-013 asks a physics reviewer to approve a pedagogically loaded simulation before it ships, and nothing automated checks this. When a draft comes back to you, the question is whether the model is right and whether the Danish reads like physics teaching.

This is not a formality. The wave draft in September 2026 gave two waves in the same medium a shared frequency and independent wavelengths, which makes them travel at different speeds — impossible in one medium, and the "beats" it produced were a drawing artefact rather than physics. It was caught during the port, not by any gate. A physicist reading it would have caught it sooner.

Tell us the reference values. The constants a tutor needs in order to mark an answer — the real efficiency, the latent heats, the medium's speed. They go in the tutor's private brief, which the student's browser never receives, under an instruction never to state them outright.

What a simulation cannot do

Worth knowing before you design around it:

- **It does not remember.** A student returning tomorrow gets their chat history but the bench reopens at its defaults. Readings that must persist belong in an activity table, not in the bench.
- **It cannot reach the internet.** No live data, no external images, no lookups. Everything ships inside the one file.
- **It does not know who the student is**, and it does not receive the activity's language yet — it takes its own default (Danish) until that is wired.

- **A constant inside it is readable by a determined student** who opens the page source. The guarantee is "not handed to you", not "cryptographically hidden". Anything that must genuinely be secret needs a different approach — say so and it will be built differently.

Who to ask

For	Ask
Handing over a draft	M or AD
Whether an idea suits a simulation or an activity element	M or AD
Physics review of someone else's draft	you, AR or JB

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